

Session 1: GenAI System Architecture & Behaviour

Duration: 120 mins

Pre-Requisites

Learners are expected to have:

- Foundational understanding of Large Language Models (LLMs)
- Prior exposure to prompting concepts
- Some familiarity with APIs and client–server architecture
- Awareness of hallucination, prompt injection, and basic LLM evaluation concepts

Session Summary

This 120-minute live session will establish the architectural foundations of GenAI systems. The focus will be on system design thinking rather than model internals.

Learners will explore the layered GenAI stack, define model versus application responsibilities, and examine orchestration, validation, governance, and monitoring. The session will contrast traditional ML paradigms with GenAI architectural patterns.

Prompting will be treated as a formal architectural layer within the system, including advanced prompt hierarchies and reasoning scaffolds. Multimodal complexity and protocol-driven tool integration will also be examined.

By the end of the session, learners will be able to diagram and explain a complete GenAI system architecture.

Learning Objectives/Outcomes

By the end of this session, learners will be able to:

- Describe the layered GenAI system stack and define component responsibilities.
- Explain why GenAI systems require orchestration beyond traditional ML pipelines.
- Design structured system prompts using hierarchical and modular approaches.

- Analyse multimodal system complexity and perceptual failure modes.
- Explain how GenAI systems integrate with external tools using protocol-driven architectures.

Brief Agenda

Learning Objective	Comments	Duration	Mode
Understand GenAI stack	Layered architecture and system boundaries	25 min	Conceptual
Compare ML and GenAI paradigms	Architectural shift and probabilistic behaviour	20 min	Conceptual
Design system prompting strategies	Prompt hierarchy and advanced patterns	30 min	Conceptual + Guided Analysis
Analyse multimodal complexity	Structured multimodal prompting and failure modes	20 min	Conceptual
Understand protocols & interoperability	Tool integration and composable systems	25 min	Conceptual + Guided Analysis

Conclusion

By the end of this session, learners will:

- Understand the structural components of GenAI systems.
- Recognise the architectural shift from deterministic ML to probabilistic orchestration.
- Treat prompting as a formal system design layer.
- Understand multimodality as an architectural complexity multiplier.
- Explain tool integration and protocol-driven interoperability.

In the next session, learners will move into model strategy, deployment, evaluation, and robustness.

Additional Reading Links

- [Generative AI | Microsoft Learn](#)
- [Best practices for prompt engineering](#)
- [What is an API? A Beginner's Guide to APIs | Postman](#)
- [What is the Model Context Protocol \(MCP\)?](#)
- [Overview of Responsible AI practices](#)